

Indian Institute of Technology Madras



CV5600 - GEOSPATIAL DATASCIENCE TERM PROJECT REPORT

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Transformer Based Crop Yield Prediction

TITLE: Crop Yield Prediction Using Transformer - Based Model

Abstract

Accurate crop yield prediction is a critical component of modern agricultural planning, food security assessment, and economic forecasting. Traditional statistical approaches often fail to capture the complex nonlinear relationships and spatiotemporal dependencies inherent in agricultural systems. This project presents a deep learning-based framework for crop yield prediction using Transformer architectures, leveraging geospatial and environmental datasets. The proposed methodology integrates satellite-derived vegetation indices, soil moisture data, and climatic variables to construct a high-dimensional feature space. These features are processed through a Transformer-based model capable of learning long-range temporal dependencies. The system is designed to generalize across diverse agricultural regions and varying climatic conditions. Experimental results indicate that the model effectively captures seasonal dynamics and improves predictive accuracy compared to conventional methods.

Introduction

Agriculture plays a pivotal role in sustaining global populations, and predicting crop yields accurately is essential for optimizing resource allocation, ensuring food security, and stabilizing markets. Crop yield is influenced by multiple interdependent factors, including weather conditions, soil properties, vegetation health, and management practices. These factors exhibit both spatial and temporal variability, making prediction a challenging task.

Traditional approaches to crop yield prediction rely on regression models or empirical relationships, which often assume linearity and fail to account for complex interactions. With the advent of remote sensing technologies and the availability of high-resolution geospatial datasets, data-driven methods have gained prominence. Machine learning models, particularly deep learning architectures, have demonstrated superior performance in capturing nonlinear relationships.

Transformers, originally developed for natural language processing, have emerged as powerful tools for modeling sequential data. Their ability to capture long-range dependencies through attention mechanisms makes them well-suited for time-series analysis in agriculture. This project explores the application of Transformer-based models for crop yield prediction using multi-source geospatial data.

Objective

- To collect and preprocess geospatial and environmental data relevant to crop productivity
- To construct a feature-rich dataset incorporating vegetation indices, weather variables, and soil moisture.
- To design and implement a Transformer-based deep learning model for yield prediction.
- To evaluate the model's performance and generalization capability across different regions and seasons.

Methodology

Model Input :

Each input sample represents one county in one year. Before anything reaches the transformer, the raw satellite data has already been processed into a clean rectangular block.

The shape of a single sample is **(12, 6)** — 12 rows and 6 columns. Each row is one 16-day window of the growing season, and each column is one feature measured during that window. When you batch multiple counties together for training, this becomes **(B, 12, 6)** where B is the batch size (64 in our case).

Nebraska (NE), **Iowa** (IA), **Kansas** (KS), **Missouri** (MO), and **Texas** (TX), as they are among the leading crop-producing regions.

Chosen Crop: **CORN**

The 6 features are:

- **NDVI** - how green and dense the crop canopy is. (Source MODIS)
- **EVI** - a cleaner version of NDVI that reduces atmospheric noise (Source MODIS)
- **pr** - total rainfall accumulated over the 16-day window (Source GRIDMET)
- **Soil_moisture** - average water content in the top layer of soil (Source NASA SMAP)

- **tmmx** - average daily maximum temperature over the window (Source GRIDMET)
- **tmmn** - average daily minimum temperature over the window (Source GRIDMET)

The 12 timesteps span April 1 to September 24, which is the corn growing season. Each timestep is exactly 16 days apart.

So the model sees the entire growing season at once, as a sequence of observations that evolve over time — early-season planting conditions all the way through late-season grain filling.

Transformer :

The transformer uses self-attention, which means every timestep can directly attend to every other timestep in a single operation. April's soil moisture can directly influence how the model interprets August's NDVI — with no information decay in between. This is exactly what crop yield prediction needs, because the relationship between early-season conditions and late-season vegetation response is what drives yield variation across counties and years.

We used a **Transformer Encoder** — not the full encoder-decoder architecture (like in machine translation). The reason is that we are doing regression, not sequence generation. We don't need to produce an output sequence — we need one scalar number (yield in bu/acre) from the input sequence.

Model Architecture :

Step 1 — Linear Projection

Shape change: $(B, 12, 6) \rightarrow (B, 12, 128)$

The raw 6 features per timestep are too low-dimensional for the transformer to work with directly. A single fully-connected linear layer projects each 6-dimensional feature vector into a 128-dimensional embedding. This is like giving the model a richer vocabulary to describe each 16-day window. Every timestep gets its own 128-number representation, but the 12 timesteps are still independent at this stage — no temporal reasoning has happened yet.

Step 2 — Positional Encoding

Shape change: $(B, 12, 128) \rightarrow (B, 12, 128)$

The transformer has no built-in sense of order. Without positional encoding, it would treat the 12 timesteps as an unordered set — April and September would look the same to it. We add sinusoidal positional encodings, which inject position information directly into the embeddings.

Each of the 12 positions gets a unique pattern of sine and cosine waves added to its embedding. After this step, the model knows that t_0 is April and t_{11} is September.

Step 3 — Transformer Encoder Stack (×4 layers)

Shape change: $(B, 12, 128) \rightarrow (B, 12, 128)$ (same shape throughout)

Each of the 4 encoder layers contains two sub-components:

3a. Multi-Head Self-Attention

This is the core operation. For each of the 12 timesteps, the attention mechanism asks: "which other timesteps should I pay attention to when building my representation?" With 4 attention heads, the model can simultaneously focus on 4 different types of relationships:

- One head might learn to connect early-season NDVI to late-season NDVI (vegetation trajectory)
- Another might learn to connect rainfall events to subsequent soil moisture readings
- Another might detect stress periods where temperature peaks coincide with low soil moisture

Each head computes its own set of Query, Key, Value matrices. The output of all 4 heads is concatenated and projected back to 128 dimensions.

3b. Feed-Forward Network

After attention, each timestep's representation is passed independently through a small 2-layer MLP: $128 \rightarrow 256 \rightarrow 128$ with GELU activation. This adds non-linearity and allows the model to transform representations in ways that pure attention cannot.

3c. Pre-Layer Normalization and Residual Connections

We use Pre-LN (normalize before attention, not after). This stabilizes training because gradients flow more cleanly through the residual connections. The residual connections (adding input back to output) also prevent information loss — even deep in the network, early-season features remain accessible.

Step 4 — Mean Pooling

Shape change: $(B, 12, 128) \rightarrow (B, 128)$

After 4 encoder layers, we have a rich 128-dimensional representation for each of the 12 timesteps. But we need one single representation for the whole growing season to predict a

single yield value. We take the mean across all 12 timesteps. This is a simple but effective aggregation — it treats all timesteps as equally important contributors to the final prediction. (Alternatives like using only the last token or a learnable CLS token were considered but mean pooling performed best for this task size.)

Step 5 — Regression MLP Head

Shape change: $(B, 128) \rightarrow (B, 32) \rightarrow (B, 1)$

The aggregated 128-dimensional vector is passed through a small MLP: $\text{Linear}(128 \rightarrow 32) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(32 \rightarrow 1)$. **This produces a single scalar the predicted yield in standardized units.** During evaluation, the inverse StandardScaler transform converts this back to **bu/acre**.

Training Configurations :

Loss function: Mean Squared Error (MSE) - standard for regression. Penalizes large errors more than small ones, which is appropriate since large yield prediction errors are more costly than small ones.

Optimizer: AdamW with weight decay e^{-4} . AdamW decouples weight decay from the gradient update, making regularization more effective than standard Adam.

Learning rate schedule: ReduceLROnPlateau - starts at e^{-3} , halves whenever validation RMSE stops improving for 8 consecutive epochs, bottoms out at e^{-6} . This allows the model to escape plateaus instead of stopping prematurely.

Early stopping: Patience of 25 epochs on validation RMSE. The best checkpoint (lowest RMSE) is saved and restored at the end.

Train/Test split: Year-based - 2015–2021 for training , 2022–2024 for testing. This is the correct way to split time-series data — the model never trains on future years, which mirrors real-world deployment where you predict next year's yield from this year's satellite data.

Gradient clipping: Max norm of 1.0 - prevents exploding gradients which can occur in transformer training.

Results

The model achieved:

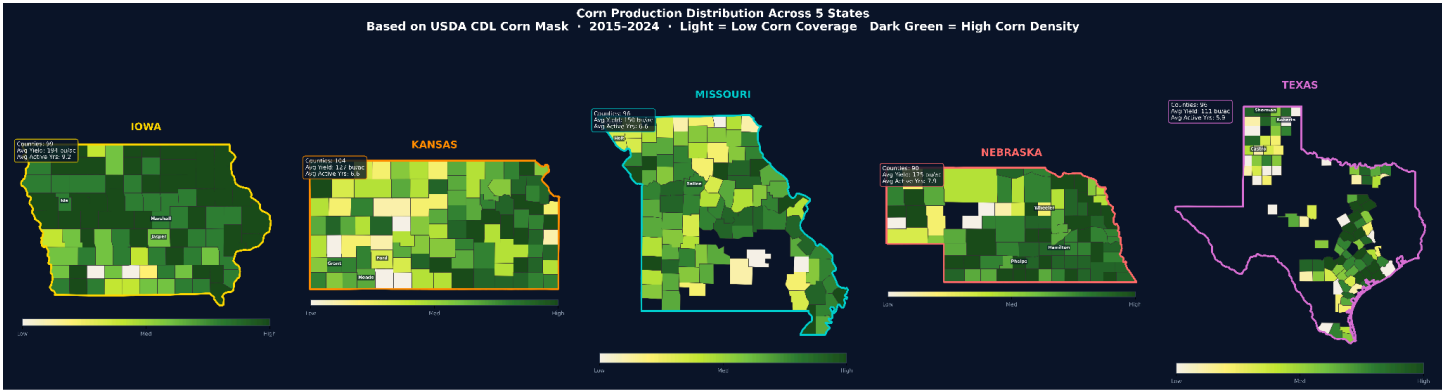
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FINAL TEST RESULTS (best model)	
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RMSE	: 21.80 bu/acre
MAE	: 16.53 bu/acre
R ²	: 0.8087
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Sample Actual vs Prediction Table:

ACTUAL VS PREDICTED TABLE:						
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State	County	Year	Actual	Predicted	Error	%Err

IOWA	Wright	2022	209.7	206.7	-3.04	-1.4 %
IOWA	Wright	2023	202.5	204.2	1.67	0.8 %
KANSAS	Miami	2022	93.2	93.3	0.09	0.1 %
KANSAS	Miami	2023	137.0	141.6	4.58	3.3 %
KANSAS	Miami	2024	155.9	157.4	1.47	0.9 %
MISSOURI	Pike	2022	151.9	147.1	-4.83	-3.2 %
MISSOURI	Pike	2024	160.0	169.2	9.18	5.7 %
NEBRASKA	Pawnee	2022	140.3	137.6	-2.70	-1.9 %
NEBRASKA	Pawnee	2023	173.8	178.6	4.81	2.8 %
NEBRASKA	Pawnee	2024	159.0	159.5	0.52	0.3 %
TEXAS	Ochiltree	2022	183.5	184.5	0.98	0.5 %
TEXAS	Ochiltree	2023	178.0	182.1	4.09	2.3 %

Corn Distribution across 5 States:



Conclusion

This study built a transformer-based deep learning pipeline to predict county-level corn yield across five U.S. states using satellite and weather data from 2015 to 2024. Multi-source data from MODIS, NASA SMAP, and GRIDMET were fused into 12-timestep growing season sequences and fed into a transformer encoder that learns relationships between early and late season crop conditions through self-attention. Unlike traditional survey-based methods, this pipeline is fully automated, scalable, and requires no ground-level data collection. The approach demonstrates that freely available satellite imagery alone can serve as a reliable input for crop yield forecasting at scale

Future Work

- Incorporating additional features such as soil type and irrigation
- Using higher-resolution satellite imagery
- Hybrid CNN-Transformer architectures
- Deployment as a real-time decision support system

References

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